A Lightweight Full Entropy TRNG With On-Chip Entropy Assurance

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***Abstract*—True random number generator (TRNG) as one essential hardware primitive is widely used in cryptography, Monte Carlo simulation, and gambling. To evaluate the security of TRNG, the entropy of the TRNG's output is usually estimated by the stochastic model in theory or measured off-chip after fabrication. However, the sufficiency of entropy is difficult to be guaranteed in practice due to the facts: 1) the inaccuracy of the model-based jitter measurement method; 2) the variations of the chip manufacturing process and operating environments (such as supply voltage and temperature); and 3) malicious attacks. In this work, we design a novel TRNG architecture with on-chip entropy assurance to properly solve practical security problems. In the design, we propose an on-chip entropy estimator for measuring independent jitter to quantify true randomness, which enables continuous monitoring of TRNG at runtime. Furthermore, with the cooperation of the proposed on-chip entropy estimator and a rational self-adaptive mechanism, the designed TRNG can steadily generate bitstreams with sufficient entropy ($\geq$ 0.999 per bit) against PVT variations. We implement the TRNG architecture in FPGAs with different technology nodes (45 and 65 nm) and SMIC 130 nm chips. Experimental results validate that the designed TRNG has an excellent performance in terms of technology independence and environmental robustness. The generated bitstreams pass the NIST SP800-22 and Diehard statistical test suites successfully without any post-processing.**

***Index Terms*—Chip implementation, high robustness, independent jitter measurement, lightweight design, on-chip entropy assurance, true random number generator (TRNG).**

I. Introduction

RANDOM number generators (RNGs) are critical building blocks for cryptographic systems. They can be broadly classified into two main categories: 1) true/nondeterministic RNGs (TRNGs) and 2) pseudo/deterministic RNGs (PRNGs). In contrast to the PRNG whose output is completely determined by the input ("seed"), TRNG harvests the entropy from the physical phenomenon, such as chaos, thermal noise, and jitter. Hence, if even the adversary knows the details about the design of a TRNG, including schematic, algorithm, parameters, etc., he/she cannot predict the output with high confidence. Thus, the security is crucial for evaluating the quality of TRNGs.

To evaluate the security of TRNGs, entropy is usually employed. Both the ISO/IEC 18031 [1] and AIS 31 [2] recommend adopting *entropy estimation* (i.e., white-box test) to comprehensively guide the design and validation of TRNGs. The theoretical entropy estimation commonly needs to establish a stochastic model which is based on an assumption of entropy source (i.e., physical phenomena) behavior. For instance, the researchers have presented some representative models through the evolution of the time [3]–[5] and the phase [6], [7] for the oscillator-based TRNG, whose structure is a typical and frequently used type in TRNG designs. Due to the extensiveness and versatility of oscillator-based TRNGs, we take this type of TRNG as our study target in this article.

Actually, only relying on the theoretical entropy estimation is not enough to guarantee the sufficiency of the entropy for running TRNGs. In the oscillator-based TRNGs, the source of randomness is the jitter[1] caused by electronic noises. In general, stochastic models are built before implementation, and the assumptions in the existing models are usually too ideal and not completely consistent with the reality. For example, Lundberg [9] described that the noise behavior of bulk CMOS devices is dominated primarily by two noise sources: 1) the thermal/white noise, which is a nonautocorrelated noise and 2) the correlated noise, which is an autocorrelated noise. In an implementation, deterministic perturbation originating from power supplies or the imbalance of duty cycles could reduce the quality of generated bit sequences [4], [5]. Thus, an on-chip entropy estimator with the capability of continuously monitoring the entropy of TRNGs at runtime is desired after deployment. For the entropy estimator, the amount of entropy is commonly reflected by calculating a measurable statistic that is selected based on the stochastic model. However, because the aforementioned undesirable factors are often introduced into the measurement, early entropy estimators are likely to be inaccurate for the quantification of true randomness, e.g., [5] and [10]. Subsequent schemes focused on how to

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[1]Jitter is defined as the short-term variations of a digital signal's significant instants from their ideal positions in time [8].

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avoid the influence of these adverse factors, such as [11]–[14]. Compared to offline computation [11], measuring the amount of jitter by the TRNG itself is preferred [12]–[14], but the consumed resource is a concern. As the oscillators only occupy a very small area and are suitable for constrained scenarios, a large measurement circuit is obviously unacceptable. Therefore, how to design *a lightweight entropy estimator* that can accurately measure the true randomness contained in a running TRNG online is *the first challenge*.

Furthermore, relying solely on the entropy estimator is still not enough, because the quality of the TRNG's output has not been improved as needed. The entropy source of running TRNGs is susceptible to the variation of physical conditions: difference of manufacturing technologies, environmental fluctuations, component aging, etc., which always influences the quality of generated data. For the oscillator-based TRNG, the jitter will continuously accumulate with the increase of sampling interval, so that the entropy will increase, too. However, in actual manufacturing, the sampling interval of generators is generally preset to a fixed value according to the worst-case result of simulation experiments, while this practice has the following security risks.

1) *Risk 1:* Presently, the sampling interval is preset to a fixed value based on the limiting cases (e.g., SS/FF corners, S: slow, F: fast) provided by simulation experiments, but the existing circuit simulators (such as Synopsys HSPICE and NanoSim) cannot precisely simulate the evolution process of noises in TRNG chips. Thus, there is still a gap between the simulation results and the noise characteristics in real chips. Actually, the noise parameters of a chip itself can only be acquired via experimental measurements after the chip has been already fabricated. Furthermore, chips with different technology nodes (e.g., 45/65/130 nm) have different noise parameters. The noise parameters of different chips under the same technology node may also be different due to the gap of chip manufacturing, and this phenomenon even occurs between different chips using the same process corner in the same batch. In view of the above uncertain factors, the amount of jitter accumulated in each chip is likely to be different under the fixed sampling interval, and the "unqualified" chips will inevitably lead to a waste of development investment.
2) *Risk 2:* The entropy source of TRNG chips is sensitive to the variations of environmental conditions [12], [15], [16], e.g., the fluctuations of supply voltage and temperature. These prior works demonstrated that attackers could manipulate voltage and temperature to bias the TRNG's output. Moreover, a seemingly perfect TRNG may be vulnerable to malicious or new attacks in the future, such as the frequency injection attack [17]. This leads to that the TRNG with a fixed sampling interval is not robust enough in the face of the environmental fluctuations and attacks, and the quality of the output is affected.

In the literature, several studies have presented approaches to improve the robustness of the oscillator-based TRNGs, through compressing raw data based on the amount of "0/1" transitions [18], reforming the structures of oscillators [19], [20], calibrating duty cycle [21], or the frequency of oscillating signals [22], etc. However, these designs are not based on entropy that represents the inherent randomness of the TRNGs, thus cannot guarantee the sufficiency of entropy at runtime. How to design a *self-adjustment mechanism* that is based on the evaluation of the entropy estimator to ensure the practical security of TRNGs is *the second challenge*.

In this work, we tackle the aforementioned two challenges and design a *practical full entropy*[2] TRNG (FE-TRNG) architecture. An on-chip entropy assurance module is embedded in the FE-TRNG to guarantee the sufficiency of entropy for the generator's output under various manufacturing technology nodes, operating environments, etc.

The contributions for this work can be summarized as follows.

1) We provide a new measurement method for the *independent jitter* due to white noise, which can quantify the amount of true randomness in the output generated by the oscillator-based TRNG. Furthermore, this measurement can be easily implemented in logic devices as an entropy estimator to monitor the quality of the TRNGs online with low overhead, which can be applied to low-power devices, such as IoT devices.
2) We propose an architecture of FE-TRNG with embedded entropy assurance for the first time. The lightweight entropy assurance is achieved via the cooperation between an *entropy estimator* and a *self-adaptive mechanism*. The entropy estimator based on our proposed measurement is utilized to continuously monitor the security of running TRNGs. We further present the first self-adaptive mechanism that adjusts the design parameters automatically in light of the result provided by the proposed entropy estimator. In this way, our designed TRNG is theoretically robust against the variation of physical conditions, and stably outputs bitstreams with sufficient entropy. Furthermore, the concise structure of the entropy assurance can be used as an elementary module employed in oscillator-based TRNGs.
3) We implement this FE-TRNG architecture in FPGAs with different technology nodes (45 and 65 nm) and SMIC 130-nm process to verify the practical security. As far as we know, this is the first chip prototype that integrates an on-chip entropy assurance module into the oscillator-based TRNG. Our TRNG architecture is *highly robust*, and can generate high-quality sequences with sufficient entropy ($\geq$ 0.999 per bit) under different manufacturing technologies and environmental conditions (voltage: 0.6–1.8 V and temperature: -10 °C–40 °C). The output sequences without post-processing successfully pass the verification of the NIST SP800-22 and Diehard test suites under these physical conditions. The area and power consumption of the

[2]Full entropy is defined as the bit-rate entropy (i.e., average Shannon entropy per internal random bit) is infinitely close to 1 per bit. The design goal in this work is that the entropy is not lower than 0.999 per bit, which meets the requirement ($>$ 0.997 per bit) in the AIS 31.

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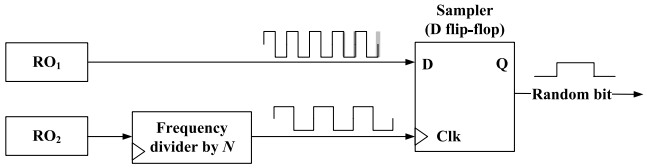

Fig. 1. Conventional structure of oscillator-based TRNGs.

ASIC prototype are about 8308 μm^2 and 0.246 mW with 1.2-V supply voltage. The generation speed is about 10 kb/s, and it can be applied in security-critical applications without the requirement of high speed, such as the seed generation for PRNGs. The analysis of experimental results also clearly reveals that several practical factors indeed affect the security of TRNGs, and emphasizes the necessity of the on-chip entropy assurance.

The remainder of this article is organized as follows. In Section II, we introduce the related work for the entropy estimation of oscillator-based TRNGs and the influence factors on entropy source in practice. An architecture of lightweight FE-TRNGs with on-chip entropy assurance is proposed in Section III. In Sections IV and V, to verify the robustness of this architecture under a variety of physical conditions, the designed TRNG is implemented in FPGAs with different technology nodes and SIMC 130 nm ASIC chips, respectively. We conclude the work in Section VI.

II. Preliminary and Related Work

The oscillator-based TRNG is generally employed because of its nice properties of the elegant and scalable structure, and it usually has a stochastic model. The recent related work on this type of TRNGs is introduced in this section, including entropy estimations and influence factors on practical security.

Fig. 1 shows a conventional design of oscillator-based TRNGs, whose randomness is derived from the instability (i.e., jitter) of oscillating signals caused by the phase noise in the circuit. The entropy source consists of two oscillators that generally exist in the form of ring oscillators (ROs). The output (oscillating signal) of one ring oscillator (RO_2) determines the events of sampling the output of the other one (RO_1) in a sampling unit, such as a D flip-flop. The sampling frequency is divided into N. The parameter N is important because it makes it possible to increase the sampling interval and thus accumulate the jitter over a longer period to achieve sufficient entropy for TRNG outputs.

A. Existing Stochastic Models for the Entropy Estimation

The stochastic model is based on the assumption of physical analog phenomena (like the jitter), and describes the principle of random number generation by mathematical notations. The entropy contained in TRNGs can be calculated theoretically via taking the value of security parameters into the model. The approaches of entropy estimation are wide researched through the evolution of time and phase.

Killmann and Schindler [3] proposed a general model for common TRNG designs with concise structure. For specific TRNG designs, the models which focus on the conventional RO-based TRNG were investigated using phase [6], [23] and timing [4], [5], [11], [12], [14] approaches. The entropy theory for multibit random number generation also has several achievements, such as the entropy estimation for ADC sampling-based TRNGs [7]. The models for some other types of oscillator-based TRNGs were presented in [24]–[26]. For the TRNGs based on metastability [27], chaos [28], [29], and complex feedback behaviors [30], unfortunately, it seems infeasible now to propose a stochastic model.

In particular, Ma *et al.* [5] pointed out a security parameter q to reflect the amount of entropy, and theoretically provided the required q corresponding to sufficient entropy (> 0.9999 per bit) under different design parameters. Moreover, according to the renewal process theory, they found that the standard variance of counting results is approximate to q with the consecutive counting mode. Furthermore, Zhu *et al.* [31] investigated the calculation error between the standard variance of counting results and the approximated q, and gave a more accurate estimation for the jitter.

B. Recent Work on the Measurement for Independent Jitter

Jitter measurement is a widely used method for entropy estimation in practice. This method can reflect the amount of entropy contained in TRNGs directly because the quality (e.g., the spectrum) and size of the jitter determine the statistical quality and unpredictability of the generated data.

In previous work, at first, the jitter measurement was realized by measuring the variance of counting results which are the number of edges on the sampled signal in one sampling interval, such as the research in [5] and [10]. However, some adverse factors other than the independent jitter contributed by white noise will be introduced in the measurement process, and resulting in inaccurate estimation of the true amount of entropy. For instance, according to the analysis [11], [32], [33] on the relationship between jitter and electronic noise, the random jitter is not only due to the white noise, i.e., independent jitter but the correlated noise impacts on the jitter, i.e., correlated jitter. This makes the counting result and even the output of TRNGs correlate with the adjacent one because of the existence of correlated noise (embodied at low frequency), which will lead to an overestimation of the entropy in the measurements.

Thus, several studies focused on the measurement for independent jitter to quantify the amount of true randomness. Assessing the contribution of white and correlated noises to the randomness of TRNGs is a difficult task. Haddad *et al.* [11] analyzed the influence of correlated noise on jitter measurement and provided an offline method to measure the independent jitter using the curve fitting of the variance of jitter. However, this approach is not suitable for online monitoring of the change of entropy. Fischer and Lubicz [12] proposed a precise method to measure independent jitter online, and the sampling interval was recommended to be set in a specific range to avoid the influence of correlated noise on jitter. Yang *et al.* [13] utilized the carry chain primitives and differential measurement to quantify the jitter accumulation under the white noise. While both of these methods used

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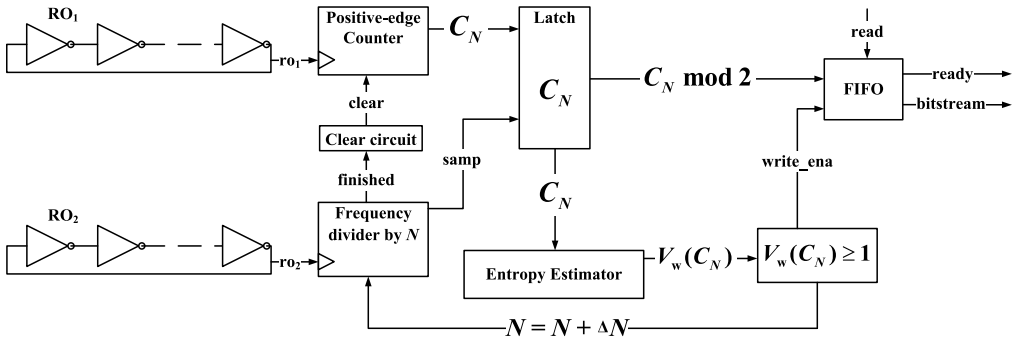

Fig. 2. Architecture of FE-TRNG with embedded entropy assurance.

a mass of computing resources in exchange for high accuracy or high speed. Furthermore, Allini *et al.* [14] implemented a measurement based on Allan variance to continuously monitor the source of randomness. The Allan variance is insensitive to correlated noise so that the independent jitter can be easily measured, and this method consumes fewer computing resources. In contrast to these existing studies (such as [11]–[14]), we propose another lightweight measurement approach and a self-adjustment mechanism, as well as the cooperative working strategy of the two components to achieve entropy assurance for TRNGs at runtime, and design an entire TRNG architecture which is implemented and verified in ASICs with lots of efforts. This is also the first ASIC chip development achievement that integrates an on-chip entropy assurance module into the oscillator-based TRNG.

C. Influence Factors on the Security of TRNGs at Runtime

In reality, many factors can influence the entropy source of oscillator-based TRNGs. For one thing, the factors are deterministic perturbation, imbalanced duty cycles, etc. Ma *et al.* [5] had shown that the deterministic perturbations make the generated bit sequence appear to be "random" and easily pass statistical tests, but they also pointed out that the seemingly random sequence actually has a vulnerability that makes it possible to predict the sequence. Amaki *et al.* [4] experimentally found that an unbalanced duty cycle limits the randomness of a TRNG.

For another thing, the factors are the difference in chip manufacturing technologies/processes, the fluctuations of supply voltage, and environmental temperature (PVT). For chip manufacturing, Rahman *et al.* [19] analyzed multiple ROs structure with different processes. With the same voltage and temperature, they pointed out that the variation of technologies influenced the amount of randomness resulting in low TRNG quality after the comparison of experimental results on the FPGAs with technology nodes 45, 90, and 130 nm. For the operating environment, Santoro *et al.* [34] researched the influence of temperature on the randomness of TRNGs for various structures: multiple ROs TRNG [35], metastable RO TRNG [15], and Galois RO TRNG [36]. They pointed out that the variation of environmental temperature causes an obvious change of the randomness. Some other studies [12], [15], [19] also focused on the same issue and concluded the similar results.

Besides, Markettos and Moore [17] provided a frequency injection attack on the multiple ROs TRNG to give rise to the frequency interlocking phenomenon for ROs, which leads to a significant reduction of the randomness of TRNG outputs.

III. Design of Full Entropy TRNG With Embedded Entropy Assurance

A. Architecture Description

On the basis of the conventional oscillator-based structure introduced in Section II, we design an architecture of the FE-TRNG in this section. Fig. 2 illustrates this architecture design that includes three function modules: 1) entropy source and extractor; 2) entropy estimator; and 3) self-adaptive mechanism and output control unit.

The entropy source of this FE-TRNG is composed of two ROs (denoted by RO_1 and RO_2). The counter records the number of positive edges of the sampled signal ro_1, and the frequency of the sampling signal ro_2 is divided by parameter N. The value of the counter within the N cycles of ro_2 will be stored in a latch, i.e., C_N. The design parameter N is set to be *adjustable* so that it is able to control the accumulation of (independent) jitter. After repeating this counting process, the entropy estimator calculates a measured statistic $V_w(C_N)$ to evaluate the contribution of *white noise* to randomness. The self-adaptive mechanism is used to judge whether the measured result meets the condition of full entropy, and to guarantee the property of full entropy for the generated data through the adjustment of the sampling interval. Note that, according to the theoretical results in [5], the entropy of the output bitstream obtained by C_N mod 2 can be sufficient if the variance of C_N is large enough with the help of jitter. Next, we will further expound the function and the working principle of each module in the following sections.

B. Entropy Source and Extractor

Deterministic perturbation introduces the deterministic factor into the result of entropy extraction and even the outputs of TRNGs, which results in a sharp decline of the difficulty to predict the generated bit [5], [6]. Besides, it can be manipulated by the adversary outside [17].

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A solution [6] using dual oscillators was proposed to eliminate the influence of deterministic perturbation. We optimize the previous method in the implementation scheme. In our entropy source, it consists of two *customized* ROs with the same number of inverters and the same topology. The dual ROs are placed at an adjacent position in a chip and have the same routing process when implemented. As we know, deterministic perturbation is produced by the power supply on a benchmark platform, and its influence is global. Due to the same perturbation on the dual ROs, the output generated by the aforementioned one RO sampling the other one does not contain a deterministic component in this way.

In the entropy extraction, we adopt one *single-edge* counter to count the number of positive edges (namely, the whole periods) of the sampled oscillating signal during each sampling interval (N periods of the signal ro_2), which can avoid the influence of the imbalanced duty cycle on generated bit sequences. After each counting process finishes, the counting result C_N is stored into a latch and the counter is cleared to start the next counting process. The C_N will be delivered to the next module to estimate the entropy in real time. The clear signal is produced through the clear circuit which is driven by both of the oscillating signals. In order to count the accurate periods in each sampling interval, we preset the number of positive edges to execute the storage and clear operations (collectively called clear process). For instance, assuming that the numbers of positive edges are N_1 and N_2 for clear and counting processes, respectively, then the number of positive edges is $N_1 + N_2$ for this corresponding sampling interval.

C. Entropy Estimator for Independent Jitter Measurement

For a running TRNG, in addition to global influence factors, the correlated noise is likely to bring security risks. We expect that the accumulation of the entropy (i.e., true randomness) is sufficient, so we provide an embedded measurement for independent jitter, which can be implemented easily in hardware and monitor the accumulation of independent jitter in TRNGs continuously.

1) Embedded Measurement for the Independent Jitter: For the conventional oscillator-based TRNG, we focus on a random process which is the accumulation of random jitter in any N ($N \in \mathbb{Z}+$) periods of oscillating signal and compute the variance of the jitter. This random process $(r_N(t_i))_{i\in\mathbb{Z}+}$ is defined as follows:

$$r_N(t_i) = \sum_{j=0}^{N-1} J\big(t_{i+j}\big)$$

where variable $J(t_k) = T_k - (1/f_0)$ represents the period jitter of the kth period in the signal, and f_0 is the nominal mean operating frequency. Thus, the variance is represented by

$$\sigma_N^2 = \mathrm{Var}(r_N(t_i)) = \mathrm{Var}\left(\sum_{j=0}^{N-1} J\big(t_{i+j}\big)\right).$$

If $\{J(t_{i+j})\}_{j=0,\ldots,N-1}$ are N mutually independent realizations, then

$$\sigma_N^2 = \sum_{j=0}^{N-1} \mathrm{Var}\big(J(t_{i+j})\big).$$

Furthermore, because the random process of jitter is stationary, namely, $\{J(t_{i+j})\}$ obey the same probability distribution, they have the same variance: for $\forall j \in \{0, 1, \ldots, N-1\}$, $\mathrm{Var}(J(t_{i+j})) = \sigma^2$. As a result

$$\sigma_N^2 = N \cdot \sigma^2. \tag{1}$$

Formula (1) means that if N consecutive jitter realizations are mutually independent in signal, the variance of accumulated jitter is a linear function of N.

The counting process in entropy extraction is usually consecutive for the conventional oscillator-based structure, i.e., oscillators are free running. According to the analysis [11], [32] of the relationship between electronic noise and jitter in oscillators, we can obtain (2) that the random jitter is mainly influenced by both white noise and flicker noise (the main component of the correlated noise). It is noticed that (2) is an approximate result because the modeling of correlated noise is too complicated to be computed analytically, and we do not focus on the nondominant components in the correlated noise

$$\sigma_N^2 = a \cdot N + b \cdot N^2. \tag{2}$$

a and b are nominal coefficients whose values do not need to be concerned. In this expression, we point out the following.

1) $a \cdot N$*:* The contribution of white noise on random jitter, noted by σ_{Nw}^2.
2) $b \cdot N^2$*:* The influence of flicker noise on random jitter.

In fact, unlike oscilloscopes, it is very difficult (even impossible) to directly estimate the size of independent jitter in the circuit. We intend to estimate the entropy (true randomness) indirectly by measuring the variance of counting results under the effect of white noise, denoted by $V_w(C_N)$. As we investigated the previous work, the stochastic model in [5] is applicable to the FE-TRNG we design because of the same working principle for random number generation. Through the model, Ma *et al.* provided a concise jitter measurement that counts both the number of positive and negative edges in a single sampling interval, and approximates the variance of the counting result (C_N) to the variance of jitter under the sampling interval. Furthermore, Zhu *et al.* [31] presented a more accurate estimation for jitter via eliminating the quantization error and the effect of waiting time in Ma's method, namely,

$$V(C_N) \approx \sigma_N^2 \cdot f_0^2 + \frac{1}{6}. \tag{3}$$

Based on (2) and (3), we can obtain the desired $V_w(C_N)$ via solving an equation set as follows (we select a pair of the multiplied number of periods N and $2N$ for convenience):

$$\begin{cases} V(C_N) = \big(a \cdot N + b \cdot N^2\big) \cdot f_0^2 + \dfrac{1}{6} \\ V(C_{2N}) = 2 \cdot \big(a \cdot N + 2 \cdot b \cdot N^2\big) \cdot f_0^2 + \dfrac{1}{6}. \end{cases} \tag{4}$$

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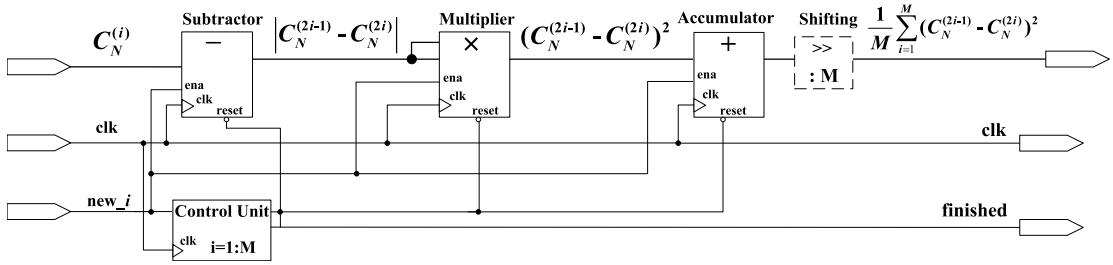

Fig. 3. Circuit structure of the embedded entropy estimator.

From (4), we have

$$V_w(C_N) = \sigma_{Nw}^2 \cdot f_0^2 = \frac{4 \cdot V(C_N) - V(C_{2N})}{2} - \frac{1}{4}. \quad (5)$$

In consideration of a characteristic of the counting results with the consecutive counting mode, i.e., for an arbitrary $i \in \mathbb{Z}+$, $C_{2N}^{(i)} = C_N^{(2i-1)} + C_N^{(2i)}$; and the calculation formula of variance $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$, we finally achieve the analytic expression for $V_w(C_N)$ as

$$V_w(C_N) = \frac{1}{2M} \sum_{i=1}^{M} \left(C_N^{(2i-1)} - C_N^{(2i)} \right)^2 - \frac{1}{4}. \quad (6)$$

2) Working Principle of the Embedded Entropy Estimator: The *entropy estimator* is implemented based on (6) to measure the $V_w(C_N)+(1/4)$. In order to elaborate on the operation process of the entropy estimator in circuitry, the structure of this measurement circuit is depicted in Fig. 3, and the procedure description is listed as follows.

1) *Step 1:* Receiving counting results $C_N^{(i)}$, we calculate the difference once a pair of adjacent counting results $C_N^{(2i-1)}$ and $C_N^{(2i)}$ are delivered into the entropy estimator, and the pair of counting results are also nonoverlapped with the previous and later, which corresponds to the operation of the subtractor.
2) *Step 2:* Calculating the squared value of the above-calculated difference, which corresponds to the operation of the multiplier.
3) *Step 3:* Accumulating the above calculated squared values with M times, where M is the preset sample size for the calculation of $V_w(C_N)$, which corresponds to the operation of the accumulator.
4) *Step 4:* Executing the arithmetic right shift of the sum produced in step 3, which corresponds to the operation of right shift with the $\log_2(M)$ bit.

Following the above procedure, we obtain the measured result $V_w(C_N)+(1/4)$. It is obvious that the proposed measurement circuit requires a small number of computing resources, which guarantees the conciseness of the entire FE-TRNG. In the circuit implementation, most of the operations are costless addition and subtraction for counting results. Moreover, because the values of the adjacent counting results are very close, the bit width of data is very small (less than 5 bits in reality) after the subtraction for the $C_N^{(2i-1)}$ and $C_N^{(2i)}$, so that the computation of this multiplier is easy. In addition, the operation of $(1/M)$ can be implemented through executing the right shift or comparing the corresponding high-order bits to the preset threshold value for the judgment of full entropy in the next module. We recommend the divisor M be expressed as the form of 2^x to facilitate the realization of the shift operation or the direct comparison. Therefore, our proposed calculation method for independent jitter is applicable to hardware implementation and consumes a small number of computing resources.

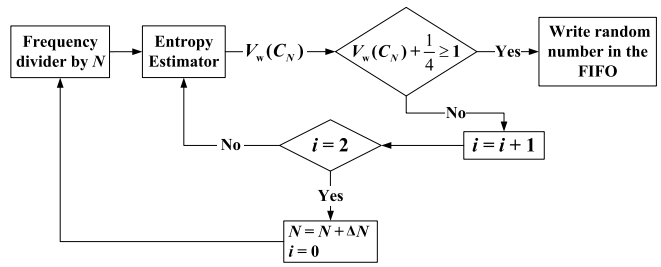

Fig. 4. Self-adaptive mechanism.

D. Self-Adaptive Mechanism and Output Control Unit

From the entropy estimator, we can continuously measure the amount of entropy for running TRNGs. The *self-adaptive mechanism* is activated to judge the sufficiency of entropy once a new $V_w(C_N)$ is produced. We expect that the bit-rate entropy of the output generated by the FE-TRNG is not lower than 0.999 (threshold value for the sufficiency of entropy in this work), which is higher than the 0.997 per bit requested in the AIS 31 [2]. According to the calculation formula of theoretical entropy given in [23], we obtain that the required quality factor q is 0.7901 in the worst case for 0.999 per bit, where $q = \sqrt{N} \cdot \sigma \cdot f_0$ is the quality factor introduced in [5]. Note that the entropy can surely become higher (closer to 1) with the growth of q. Based on (5), we can further achieve the following approximation for $V_w(C_N)$:

$$V_w(C_N) + \frac{1}{4} \approx q^2 + \frac{1}{4}.$$

Thus, the measured value $V_w(C_N) + (1/4)$ in the proposed entropy estimator should be equal to 0.8743 corresponding to $q = 0.7901$. For the sake of computing efficiency on hardware, we preset the threshold of measurement parameter in the self-adaptive mechanism to be 1, namely, $V_w(C_N) + (1/4) \geq 1$, to check whether the entropy of random bitstreams derived from the C_N mod 2 is sufficient (≥ 0.999 per bit) or not, and the real q at this time is 0.8660 (> 0.7901).

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The self-adaptive mechanism (Fig. 4) is described as follows.

1) If the condition of full entropy is satisfied, namely, $V_w(C_N) + (1/4) \geq 1$, the generated random bit from this point forward is allowed to be stored in the output device FIFO until the next judgment result is produced. The stored random bits will wait for the read command to output.
2) If the condition of full entropy is unsatisfied, namely, $V_w(C_N)+(1/4) < 1$, a preset value ΔN will be added to the parameter of sampling interval N in order to accumulate more independent jitter. While the generated random bit from this point forward is not allowed to be stored until the status of full entropy is reached (N also stops growing when the entropy is sufficient).

As a result, the output has the property of full entropy in theory with the *adjustable* sampling interval in the cases of different technologies, change of environmental conditions, etc. Furthermore, in order to ensure that the generation speed can be kept stable after reaching the status of full entropy, the operation of increasing the sampling interval will not be executed until the condition of full entropy is unsatisfied *successively twice* in implementation.

Collaboration With Other Health Tests: The designed self-adaptive structure for entropy can also be treated as one type of health test for the TRNG. That is, if the randomness degrades to the preset threshold, the impact will be reflected in the measured result of the entropy estimator, and the output is suspended until the adjustment is completed. Although the entropy measurement and the adjustment process are a bit slow, it is adequate to cope with the nonsudden change of environments (e.g., supply voltage and temperature), aging, etc. However, in order to cope with more complex attacks (such as voltage attacks and frequency injection attacks), more efficient and agile health test methods shall be used to detect the attacks as soon as possible, such as total failure test (tot test). Here, we present the design idea based on our proposed measurement circuit to perform tot test.

Generally, the voltage or frequency injection attack manipulates the oscillating frequencies of oscillators, which makes the frequencies much complex or very consistent with the manipulation frequency, such as ring-locking. In practice, without the injection of external frequency or the intentional placement, one free-running RO is hard to be interlocked with the other. The frequencies of the two ROs in our entropy source are somewhat different, and research [16] experimentally confirmed that a phase drift between oscillators is observed if the frequencies are not closely matched. Furthermore, careful routing and rings isolation in the implementation can avoid interlock [16], [37], so the phenomenon of interlocking is not observed in our experiments. Even if the interlock occurs due to an external attack, we also provide the following tot test method to deal with it. We can design a monitor for ΔC, which represents how many cycles of RO_1 within the ΔN cycles of RO_2. If ΔC keeps unchanged for a long time (such as 256 times), meaning that the randomness is lacked perhaps due to frequency locking [37], or if ΔC exceeds the expected range (such as $\Delta N \pm 20\%$), meaning that the oscillators are abnormal perhaps due to voltage attacks, the monitor for tot test will output an alarm signal and other outputs of the TRNG will be forbidden. In this way, we can implement a fast response against the failure of the entropy source.

TABLE I
AREA COMPARISON OF THE ENTROPY ESTIMATORS IMPLEMENTED IN FPGAS

Method	Device	Area	
		ALM/Regs	DSPs
Haddad *et al.* [11]	Cyclone V	119/160	2
Fischer and Lubicz [12]	Cyclone V	169/200	4
Yang *et al.* [13]	Cyclone IV	$\approx$ 1024/1024	0
	Spartan-6	$\approx$ 256/1024 (Slice/Regs)	0
Allini *et al.* [14]	Cyclone V	49/117	1
This work	Cyclone V	57/55 (38/55)	0 (1)
	Virtex-5	25/55 (Slice/Regs)	0

IV. FPGA IMPLEMENTATION OF THE FE-TRNG ARCHITECTURE

In this section, we implement our proposed FE-TRNG architecture in FPGAs to preliminarily validate the effectiveness of the design scheme. Our architecture has strong applicability to FPGA and ASIC platforms, although the principles of implementing digital logic are different for them. Besides, the experimental results on the custom-built ASIC chip more faithfully reflect the essence of the circuit. Thus, we finished various kinds of robustness experiments on the ASIC chip case study. Meanwhile, considering that the development of ASIC chips with different technology nodes requires a huge investment, we only carry out the verification experiments of functionality and technology independence on FPGAs prudently in view of the programmability of FPGAs.

A. Parameter Setting

The preset parameters of this FE-TRNG in FPGAs with different technologies are initial sampling interval $N_0 = 1280$, increment of sampling interval $\Delta N = 256$, and a divisor of the right shift in entropy estimator $M = 2^{14}$, where $M = 2^{14}$ is obtained by a tradeoff between computation complexity and stability of measurement, and the variance of the measured results is less than 1‰. The judgment of full entropy in the self-adaptive mechanism is $V_w(C_N) + (1/4) \geq 1$.

B. Implementation Results of Entropy Estimators

We evaluate the resource consumption of recent entropy estimators that are utilized to measure independent jitter. The area results are obtained from the Quartus software, and we also give the area obtained from the ISE software for our estimator. The results are presented in Table I.

We observe that the resource consumption of the lightweight entropy estimator proposed in this work is significantly lower than that of the estimators provided in all other works. Note that the results in parentheses (38 ALM/55 Regs and 1 DSP) are for ease of comparison with other works, and the DSP block for multiplication is not required actually because the bit width of the multiplier factor can be controlled to be very small (i.e., less than 5 bits).

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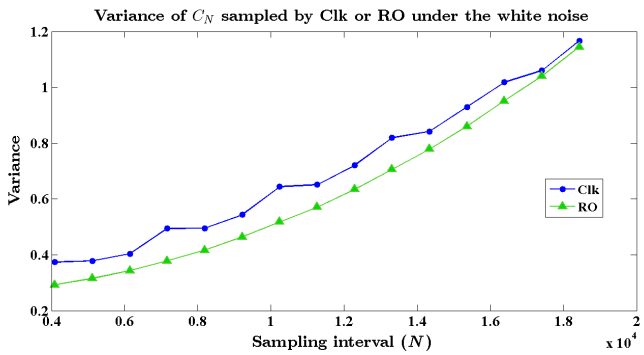

Fig. 5. Results of the proposed measurement with stable clock versus dual ROs.

Specifically, the work [11], [12] implemented the original calculation formula of variance in their measurement circuits, which requires lots of storage resources and multipliers. The differential measurement [13] consists of two identical parts, and each one contains a large number of carrying chains and D flip-flops. Relatively, our measurement circuit is more lightweight, because we simplify the calculation formula of variance on the basis of the characteristics of consecutive counting mode, and reduce the number of multipliers via minimizing the operand width. We further optimize the use of intermediate variables to reduce the number of occupied registers, and the final number of registers is only 55.

C. Functional Verification

We intend to verify the following two functions to ensure that the FE-TRNG architecture can work as expected in practice, which is implemented in the Xilinx ML 505 benchmark platform featuring Virtex-5 (65 nm) FPGA. Each RO of the entropy source is composed of an AND gate and 13 inverters, the working frequency of which is approximately 196 MHz. Besides, the dual ROs are placed at an adjacent location in the FPGA and have a similar routing delay. Note that it should avoid the intertwined routing which may lead to the mutual interference of the dual ROs.

1) Eliminating the Influence of Deterministic Perturbation: We select the stable slow clock signal (like generated by crystal oscillator) and oscillating signal (generated by RO) as the sampling signal to sample another oscillating signal as the sampled signal, respectively. The working frequency of the crystal oscillator is 100 MHz. In the two cases described above, we collect the $V_w(C_N)$ calculated by (6) with different values of sampling interval N, and observe the trend of fitting curves to judge whether the influence of deterministic perturbation can be eliminated for our entropy source structure. The fitting curves of these two groups of $V_w(C_N)$ are depicted in Fig. 5 with increasing sampling intervals.

From Fig. 5, we observe the existence of deterministic perturbations which leads to a sinusoidal shape of the rising trend for the experimental data with the stable slow clock, and the perturbation also enhances the value of measured $V_w(C_N)$ compared to another curve. This is consistent with those experimental phenomena in [5] for the $V(C_N)$. Reversely, the wavy shape disappears for the experimental data with the dual oscillators. The reason is based on the fact that the influence of deterministic perturbations is global, so the effect on the dual ROs satisfies the requirement mentioned in Section III-B is almost the same. Thus, the influence can be eliminated after utilizing one oscillating signal to sample the other one.

2) Correctness Verification of the Independent Jitter Measurement: In order to verify the correctness of the proposed measurement for independent jitter, we introduce two (internal) measurement methods as follows and compare the consistency of measured results.

1) *M1:* We select the measurement proposed in [11] as the first method, which represents the type of *reset* counting process and *difference* measurement mode.[3]
2) *M2:* The second method represents the type of *consecutive* counting process and *direct* measurement mode,[4] which adopts dual free running oscillators as the entropy source.

Obviously, M2 is the foundation of the measurement proposed in this article. Therefore, if we can obtain the same measured results (i.e., the size of period jitter) from M1 and M2, it means that our proposed entropy estimation is able to accurately measure the independent jitter.

Using the dual ROs as entropy source, we plot the fitting curves of $V(C_N)$ as the function of sampling interval N with experimental results produced by these two methods, which are depicted in Fig. 6. Furthermore, the variance of counting results due to the white noise can be represented as follows according to (2) and (5):

$$V_w(C_N) = a \cdot f_0^2 \cdot N = \sigma_{Nw}^2 \cdot f_0^2$$

where $a \cdot f_0^2$ is the first-order coefficient of the fitting curve expression, and $\sigma_{Nw}^2 = 2 \cdot N \cdot \sigma^2$ for M1 or $\sigma_{Nw}^2 = N \cdot \sigma^2$ for M2. Hence, we can calculate the period jitter σ via the first-order coefficient and mean operating frequency f_0.

For M1 and M2, the relationship of the first-order coefficients is verified from Fig. 6, that is, the coefficient ($3.36 \cdot 10^{-6}$) acquired by M1 is twice the one ($1.68 \cdot 10^{-6}$) acquired by M2. We also obtain the same period jitter $\sigma = 6.6$ ps through the two first-order coefficients and $f_0 = 196$ MHz, which is also close to our observations acquired by external measurement methods, i.e., oscilloscope. Besides, we find that the accuracy of the curve fitting measurement [11] for period jitter depends to a great extent on the stability of the measured variance $V(C_N)$ and the precision of the curve fitting algorithm. Therefore, we have verified the correctness of the presented measurement for independent jitter.

D. Independence Verification for Manufacturing Technology

In order to verify the independence of this FE-TRNG architecture from manufacturing technologies, we implement

[3]Difference measurement mode is defined as the variance of counting results $C_N^{(i)}$ that is derived from the difference between the adjacent output $C_N^{(2i-1)}$ and $C_N^{(2i)}$ of positive (or negative)-edge counter.

[4]Direct measurement mode is defined as the variance of counting results $C_N^{(i)}$ that is directly derived from the output $C_N^{(i)}$ of positive (or negative)-edge counter.

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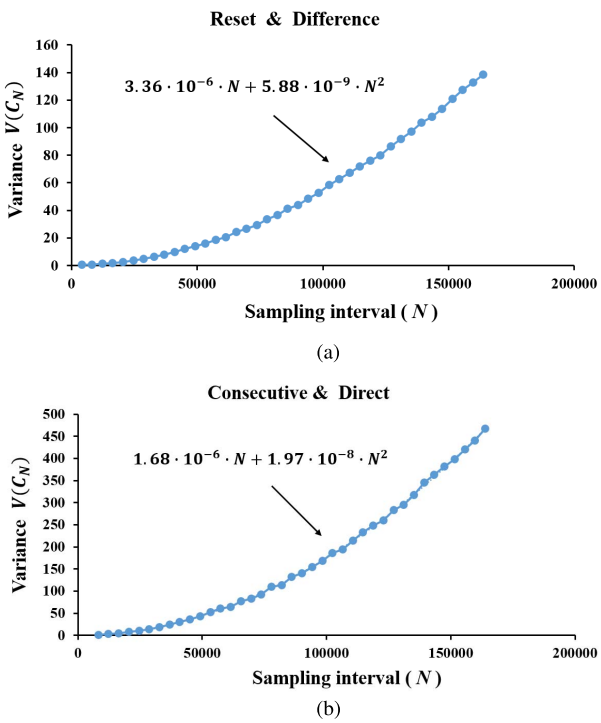

Fig. 6. Experimental results of the two measurement methods. (a) Fitting curves of $V(C_N)$ obtained by M1 with increasing N. (b) Fitting curves of $V(C_N)$ obtained by M2 with increasing N.

our design in FPGAs with two different technology nodes: 1) the Xilinx ML 505 benchmark platform featuring Virtex-5 (65 nm) FPGA and 2) Spartan-6 (45 nm) FPGA. The working frequencies of two 13-stage ROs are 196 MHz in Virtex-5 (and 123 MHz in Spartan-6), which have the same placement and routing for inverters (delay elements), and the positions of these two oscillators are as close as possible to let them be influenced by the nearly identical deterministic perturbation.

1) Experimental Result for Technology Independence: Table II shows the experimental result for the technology independence of this FE-TRNG implemented in these two FPGAs. We can observe that the variance $V_w(C_N)$ gradually increases with the growth of sampling interval from initial preset value. After achieving the status of full entropy $\left(V_w(C_N) + (1/4) \geq 1\right)$, the variation trend of $V_w(C_N)$ will become steady quickly, so the generation speed is also able to be stable. From the experimental result, the sampling interval on the Virtex-5 and Spartan-6 FPGA is $N = 17\,920$ and $N = 37\,120$, respectively, when the TRNG runs stably in the status of full entropy. That is to say, under the same security requirement, e.g., the full entropy, if the preset sampling interval is fixed with $N = 17\,920$, then the entropy of generated bit sequence from the Virtex-5 is sufficient, while the entropy is insufficient for the Spartan-6 case. Therefore, the sampling interval needs to be adjustable to ensure the high-quality output of TRNGs with a variety of technologies.

2) Statistical Randomness Test: With the implementations on different FPGA platforms, we employ the rigorous statistical test suites, including NIST SP 800-22 [38] and Diehard [39] to assess the quality of the TRNG output.

TABLE II
VERIFICATION ON TECHNOLOGY INDEPENDENCE FOR THE FE-TRNG EMBEDDED IN VIRTEX-5 (65 NM) AND SPARTAN-6 (45 NM)

Virtex-5			Spartan-6		
N	x	status	N	x	status
4096	0.2928	Insufficient	10240	0.3197	Insufficient
5120	0.3156		12288	0.3598	
...	...		...	...	
15360	0.8596		32768	0.8864	
16384	0.9508		34816	0.9594	
17408	1.0400	Sufficient (unstable)	36864	1.0179	Sufficient (unstable)
17920	1.0858	Sufficient (stable)	37120	1.0364	Sufficient (stable)

Note: $x = V_w(C_N)$, **status** represents the amount of entropy.

a) NIST SP 800-22 test suite: A 10^9-bit data with binary form is tested by the NIST test suite. The tested data consists of 1000 consecutive sequences of 10^6 bits. NIST has provided two approaches to evaluate the quality of tested sequence, including: 1) *First-Level Test:* the proportion of sequences that pass a statistical test and 2) *Second-Level Test:* the uniformity of the distribution of p-values, denoted by P_value_T. The proportion of tested sequences that passed lying in [0.981, 0.999] (the default significance level $\alpha = 0.01$), and P_value_T above 10^{-4} are considered acceptable. The test result shows that the tested data can successfully *pass* the NIST test suite *without post-processing*, and detailed records are listed in Table III. Compared to the single-edge counting, we also observe that the data cannot pass the test suite under the double-edge (the positive/negative edge) counting with the same experimental conditions due to the imbalance of duty cycles.

b) Diehard test suite: A 2^{30}-bit data is tested by the Diehard battery of tests (version 3.31.1). The significance level $\alpha = 0.01$, and the tested sequence is considered "qualified" if the calculated P-value (uniformity of p-values) is greater than 0.0001. The test result shows that all the test items are passed. Detailed records for each test item are listed in Table IV.

V. ASIC IMPLEMENTATION OF THE FE-TRNG ARCHITECTURE

In this section, we implement the designed architecture in ASIC chip to further validate the effectiveness of the FE-TRNG for practical security. The experimental verification contains the following aspects.

1) Independence from individual manufacturing difference.
2) Robustness against the changes of supply voltage and environmental temperature.

The individual manufacturing difference represents the inherent deviation among chips with the same technology, such as the frequency of ROs and the size of noise in the circuit.

A. ASIC Chip Implementation

The FE-TRNG architecture is implemented in a prototype chip under SMIC 130-nm CMOS technology, which is a purely digital circuit. The entropy source contains dual ROs that have

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TABLE III
TEST RESULTS OF THE NIST SP800-22 FOR THE ROBUSTNESS OF THE FE-TRNG AGAINST PVT VARIATIONS

Test item	Manufacturing technology (at 25 °C, 1.2 V)								Voltage (at 25 °C)				Temperature (at 1.2 V)			
	Virtex-5		Spartan-6		Chip No.1		Chip No.6		0.9 V		1.8 V		-10 °C		40 °C	
	Prop.	P_v	Prop.	P_v	Prop.	P_v	Prop.	P_v	Prop.	P_v	Prop.	P_v	Prop.	P_v	Prop.	P_v
Freq.	98.8%	0.219	98.7%	0.220	99.1%	0.554	99.5%	0.699	98.7%	0.350	99.5%	0.383	99.2%	0.834	99.6%	0.213
Block Freq.	98.9%	0.906	99.0%	0.894	99.3%	0.743	99.1%	0.086	99.4%	0.596	99.1%	0.699	99.0%	0.350	99.1%	0.911
Cum. Sums	98.9%	0.441	98.9%	0.439	99.0%	0.789	99.0%	0.534	99.3%	0.779	99.6%	0.213	98.9%	0.740	99.3%	0.350
Runs	99.0%	0.656	98.9%	0.739	99.4%	0.795	99.0%	0.658	98.5%	0.699	99.3%	0.972	98.8%	0.437	98.8%	0.804
Longest Run	98.8%	0.614	98.8%	0.593	99.3%	0.692	99.2%	0.883	99.6%	0.304	99.0%	0.658	99.1%	0.911	99.0%	0.546
Rank	98.5%	0.230	98.6%	0.205	99.5%	0.012	98.8%	0.534	99.2%	0.479	99.4%	0.514	99.4%	0.637	99.2%	0.602
FFT	98.7%	0.280	98.7%	0.243	99.2%	0.290	99.4%	0.067	99.0%	0.401	98.2%	0.554	99.3%	0.871	99.1%	0.534
Overlapping Template (OT)	98.7%	0.051	98.6%	0.133	98.6%	0.299	99.1%	0.205	99.5%	0.616	99.3%	0.304	99.0%	0.956	99.4%	0.950
Non-OT	98.9%	0.701	99.1%	0.621	99.6%	0.062	99.3%	0.213	99.1%	0.361	99.2%	0.225	99.2%	0.437	98.6%	0.740
Universal	99.0%	0.222	99.2%	0.318	99.0%	0.046	98.6%	0.016	98.3%	0.067	99.6%	0.720	99.1%	0.873	99.0%	0.816
Appr. Entropy	99.2%	0.365	99.1%	0.408	99.1%	0.017	99.6%	0.779	99.8%	0.115	99.4%	0.304	99.0%	0.739	99.3%	0.407
Random Excursions (RE)	99.7%	0.900	99.8%	0.831	99.7%	0.270	99.5%	0.111	99.7%	0.276	99.5%	0.306	99.7%	0.964	99.8%	0.834
RE Variant	99.5%	0.037	99.7%	0.098	99.8%	0.073	99.5%	0.086	99.5%	0.437	99.7%	0.451	99.5%	0.276	99.7%	0.342
Serial	98.5%	0.695	98.8%	0.752	98.8%	0.278	98.9%	0.817	99.7%	0.304	98.6%	0.103	98.5%	0.213	99.2%	0.511
Linear Complexity	99.9%	0.965	99.5%	0.932	99.0%	0.682	99.3%	0.851	99.6%	0.637	99.1%	0.534	99.5%	0.932	99.5%	0.437

Note: (1) the values of testing parameters are the default in the NIST test suite [39], (2) passing condition: Proportion (Prop.) $\geq 98.1\%$ and $P_\text{value}_T(P_v) \geq 0.0001$.

the same circuit structure, and each RO has one AND gate, one NOT gate, and some delay elements. The simulated frequency of the ROs is about 180 MHz using the tt process corner, and the practical working frequency is about 160 MHz. The reference values of voltage and temperature for the chip are 1.2 V and 25 °C during normal working. The layout of the FE-TRNG chip includes an entropy source with a dual ROs form, entropy extraction of single-edge counting, entropy assurance, and output control module.

The preset parameters of this FE-TRNG on prototype chips are initial sampling interval $N_0 = 10\,240$, increment of sampling interval $\Delta N = 256$, and divisor of right shift in entropy estimator $M = 2^{14}$. The judgment of full entropy in self-adaptive mechanism is $V_w(C_N) + (1/4) \geq 1$.

The pins of the prototype chip are listed in Table V. The input signals are VCC, GND, CLK, and ENABLE, and the output signals are Bit_Out and READY. Some test pins are reserved to validate the proposed architecture, such as measured parameter $V_w(C_N)$ and sampling interval N, and can be encapsulated on demand.

B. Chip Testing and Results

We experimentally demonstrate the influence of various physical conditions on the randomness of TRNGs, and verify whether our design has a satisfactory performance under these cases. With the benefit of the mechanism of the entropy assurance, the influence can be reflected by the change of sampling interval with different physical conditions. Actually, the physical condition has an effect on both the frequency of ROs and the sampling interval, but the change of sampling interval occupies the *dominant* compared to that of the frequency, which we will illustrate in the following result analysis.

1) Test Platform: The test platform of the FE-TRNG ASIC chip is depicted as Fig. 7. The test chip is put on a high–low-temperature test chamber to adjust and maintain a specific value of environmental temperature. An adjustable voltage-stabilized source supplies 1.2 and 3.3 V standard voltages to the test chip. In order to observe the value and frequency of output signals, the pins of output signals are connected with a logic analyzer and an oscilloscope. The bit strings generated by the FE-TRNG chip will be input to an FPGA development board, and then transmitted into a computer via serial port to conduct the statistical test.

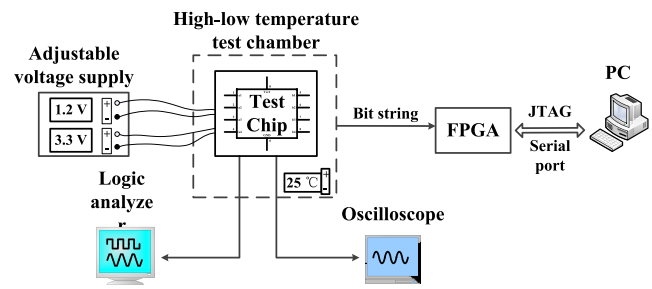

Fig. 7. Test platform for the FE-TRNG chip.

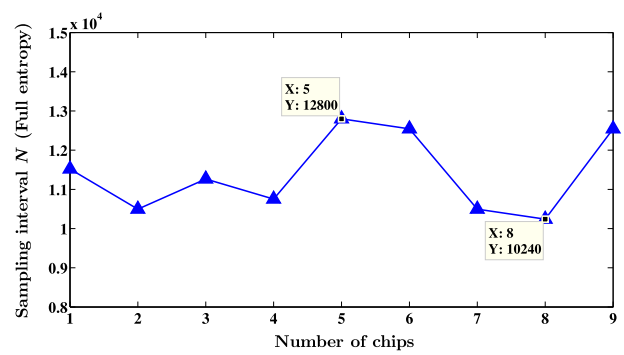

Fig. 8. Test results of the independence from individual manufacturing difference.

2) Independence Verification for Individual Manufacturing Difference: We randomly select nine test chips from the same batch with 130-nm CMOS technology to perform the independence verification for individual manufacturing differences. Under the constant environmental condition: supply voltage 1.2 V and environmental temperature 25 °C, we test each chip in turn to observe whether the test chip can reach the status of

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TABLE IV
TEST RESULT OF THE DIEHARD FOR THE ROBUSTNESS OF THE FE-TRNG AGAINST PVT VARIATIONS

Test item	**Manufacturing technology (at 25 °C, 1.2 V)**				**Voltage (at 25 °C)**		**Temperature (at 1.2 V)**	
	Virtex-5	**Spartan-6**	**Chip No.1**	**Chip No.6**	**0.9 V**	**1.8 V**	**-10 °C**	**40 °C**
	***P*-value**							
Birthday spacings	0.995	0.998	0.982	0.991	0.936	0.912	0.935	0.956
Overlapping 5-permutation	0.128	0.163	0.187	0.173	0.173	0.236	0.136	0.229
Binary rank test for 32×32 matrices	0.021	0.069	0.078	0.053	0.057	0.065	0.059	0.078
Binary rank test for 6×8 matrices	0.887	0.759	0.691	0.836	0.876	0.838	0.812	0.865
Bitstream	0.178	0.156	0.167	0.203	0.139	0.159	0.129	0.169
OPSO	0.234	0295	0.286	0.232	0.255	0.237	0.213	0.293
OQSO	0.259	0.330	0.233	0.440	0.285	0.317	0.269	0.328
DNA	0.144	0.206	0.238	0.127	0.168	0.207	0.199	0.217
Count-the-1's test on a stream of bytes	0.658	0.547	0.548	0.679	0.598	0.622	0.612	0.678
Count-the-1's test for specific bytes	0.073	0.236	0.199	0.118	0.129	0.157	0.134	0.173
Parking lot	0.353	0.484	0.390	0.468	0.532	0.569	0.483	0.582
2D spheres	0.533	0.488	0.470	0.506	0.506	0.548	0.565	0.593
3D spheres	0.317	0.258	0.325	0.289	0.333	0.318	0.298	0.346
Squeeze	0.066	0.046	0.160	0.086	0.127	0.154	0.096	0.178
Overlapping sums	0.378	0.584	0.485	0.517	0.268	0.325	0.306	0.352
Runs	0.653	0.548	0.568	0.623	0.483	0.516	0.456	0.586
Craps	0.258	0.232	0.267	0.290	0.286	0.365	0.320	0.365
Final result	Success	Success	Success	Success	Success	Success	Success	Success

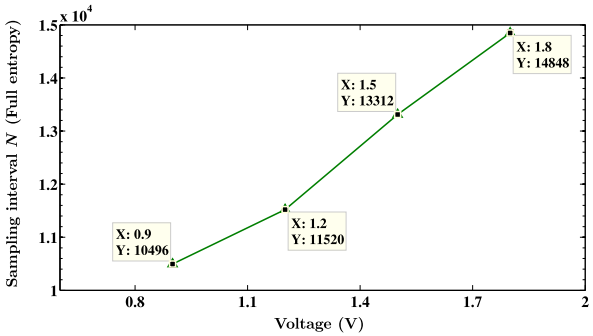

Fig. 9. Test result of the robustness for voltage variation.

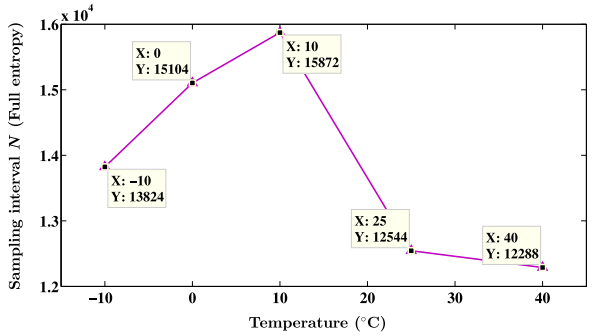

Fig. 10. Test result of the robustness for temperature variation.

full entropy with manufacturing error. When the full entropy condition is satisfied for a tested chip, we record the value of the sampling interval N and collect the generated data to take the statistical tests. Each chip needs to be restarted after it has been powered up. The experimental result is shown in Fig. 8.

In Fig. 8, the horizontal axis represents the chip number and the vertical axis represents the values of the sampling interval N under the status of full entropy. Due to the influence of individual manufacturing differences on the tested chip, the fluctuation of the frequency of ROs is nearly 5%, and that of

TABLE V
PINS OF THE DESIGNED PROTOTYPE CHIP

Pin name	**Input/Output**	**Explanation**
VCC	input	supply voltages for the TRNG chip, including 1.2 V for chip core, 3.3 V for PAD
GND	input	ground wire
CLK	input	an external input asynchronous clock (up to 100MHz) used to read the output data from the chip
ENABLE	input	an external input signal used to excite the ring oscillator, and a value of "1" means valid, and a value of "0" means invalid
Bit_Out	output	generated random bits that are in binary representation
READY	output	an output signal indicates that the output of the Bit_Out signal is random data at the moment, and a value of "1" means valid, and a value of "0" means invalid

the sampling interval is roughly 25%. Note that we select these test chips from the same batch in manufacturing to reduce the inherent difference among the chips, in other words, the fluctuation may be more than 25% for the chips from different batches. Thus, the influence of the individual manufacturing difference should be respected in practice. In addition, all of the test chips can reach the status of full entropy with different sampling intervals after powering up. Also, we collect the output generated by arbitrary two of the test chips to take the statistical tests. The test results show that the output of each tested chip can pass both the NIST SP800-22 and Diehard test suites successfully, and detailed records are listed in Tables III and IV. Therefore, the independence of the proposed FE-TRNG design from the individual manufacturing difference is validated.

3) Robustness Verification for Environmental Condition Variations: We adopt the single-variable method to perform the robustness verification for environmental condition variations. Under the variation of supply voltage or environmental temperature, we randomly select and test one or several chips in turn to observe whether the test chips can reach the status of

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TABLE VI
COMPARISON RESULTS WITH TRNG DESIGNS IMPLEMENTED IN FPGAS AND ASICS

Ref.	Source	Technology	Entropy Est.	Self-Ad.	Environment	Speed	Area	Power	Evaluation
FPGA Implementations									
[19]	Oscillator	Spartan-3E/6, Virtex-II	No	Yes	1.2 $\sim$ 1.8 (V) 0 $\sim$ 70 (^oC)	N/A	446 Slices for Spartan-3E	N/A	11 tests in SP 800-22
[41]	Oscillator	Spartan-6	Offline	No	N/A	14.3 Mbps	67 Slices	N/A	SP 800-22
[42]	Oscillator	Virtex-5	Offline	No	N/A	4.08 Mbps	109 Slices	N/A	FIPS 140-2, SP 800-22
[26]	Oscillator	Spartan-6 Cyclone V	Offline	No	N/A	1.15 Mbps 1.07 Mbps	16 LUTs+11 FFs 16 LUTs+12 FFs	N/A	SP 800-90B, AIS 31 N/A
ASIC Implementations									
[22]	Oscillator	65 nm	No	Yes	0 $\sim$ 75 (^oC)	7.5 Mbps	6670 μm^2	N/A	SP 800-22, Diehard
[21]	Oscillator	65 nm	No	Yes	0.8 $\sim$ 1.2 (V)	2 Mbps	6000 μm^2	0.13 mW	SP 800-22
[20]	Oscillator	40 nm	No	Yes	0.6 $\sim$ 0.9 (V) -40 $\sim$ 120 (^oC)	2 Mbps	836 μm^2	0.046 mW	SP 800-22
[43]	Metastable	180 nm	Offline	No	1.8 $\pm$ 0.15 (V) -20 $\sim$ 60 (^oC)	2.5 Mbps	1240.5 gates	0.27 mW	SP 800-90B, AIS 31
This work	Oscillator	Virtex-5 Spartan-6 130 nm	Online	Yes	N/A 0.9 $\sim$ 1.8 (V) -10 $\sim$ 40 (^oC)	10.9 kbps 3.31 kbps 10 kbps	72 Slices 73 Slices 8308 μm^2	0.95 mW 0.92 mW 0.246 mW	SP 800-22, Diehard

full entropy. When the full entropy condition is satisfied for a tested chip, we record the value of the sampling interval *N* and collect the generated bit sequences to take the statistical tests. After the voltage or temperature is reset, the tested chip needs to be restarted. The ranges of the change on voltage and temperature are 0.9–1.8 V (observation points: 0.9, 1.2, 1.5, and 1.8 V) and −10 °C–40 °C (observation points: −10 °C, 0 °C, 10 °C, 25 °C, and 40 °C), respectively. Note that the constant environmental temperature is 25 °C in the experiments of voltage variation, and the constant supply voltage is 1.2 V in the experiments of temperature variation. The experimental results of the robustness for voltage and temperature variations are shown in Figs. 9 and 10, respectively.

For the various values of voltage and temperature, Figs. 9 and 10 depict the different values of the sampling interval *N* under the status of full entropy. Due to the influence of environmental condition variations on the tested chip, the fluctuation of the frequency of ROs is less than 10%, and that of the sampling interval is about 30% and 40% for the experiments of voltage and temperature variations, respectively. Thus, the influence of operating environments is also noteworthy. Furthermore, we find that the tested chip can reach the status of full entropy for various values of the voltage and temperature after powering up. Also, we validate the statistical property of the output by the statistical tests under different environmental conditions. The test results show that the output of each tested chip can pass both the NIST SP800-22 and Diehard test suites successfully, and we select the boundary test points as examples to show detailed records in Tables III and IV. Therefore, the proposed FE-TRNG design has strong robustness against the environmental condition variations.

4) Comparison With Related Work on Hardware Implementations: Table VI lists recent ASIC or FPGA TRNG implementation results in the aspects of entropy source, technology, entropy estimation (Entropy Est.), self-adjustment (Self-Ad.), operating environment, generation speed, area, power consumption, and statistical randomness tests (Evaluation). These selected TRNGs involve either entropy estimation or self-adjustment technologies.

The lightweight entropy assurance module is integrated into our design, which contains a complete set of entropy monitoring and self-adaptive functions to ensure that the generator is robust enough under various physical conditions (voltage: 0.9–1.8 V, temperature: −10 °C–40 °C) during the startup and operation phases. The entropy assurance function is achieved through the cooperative work between an online entropy estimator based on a theoretical model and a self-adjustment mechanism for design parameters. In terms of area and power consumption, since the prototype chip we implemented is mainly used for the verification of the proposed security architecture, it has not been rigorously optimized for resource consumption. The area (8308 μm^2) and power consumption (0.246 mW) of our ASIC implementation are comparable to or slightly higher than those of the related work, and the entropy assurance circuit occupies about 70% of the resources required by the entire TRNG. Besides, our security architecture is designed based on the conventional structure of oscillator-based TRNG, which is convenient for modeling and integration, but the speed is relatively low due to the small jitter. In order to ensure the sufficiency of the entropy for output, we make a concession on speed, and the speed can be greatly improved through increasing the probability of capturing jitter [35] or the sensitivity to jitter [7]. Additionally, with the help of the entropy assurance, the generated high-quality bitstreams can successfully pass the NIST SP 800-22 and Diehard test suites under a variety of test conditions in this work, and the bit-rate entropy is not less than 0.999 in theory.

VI. CONCLUSION

In this article, an FE-TRNG with on-chip entropy assurance is designed and implemented for the first time to solve practical security problems. We implement the design in FPGAs with different technology nodes (45 and 65 nm) and SMIC 130 nm ASIC chips to accomplish the verification of practical security. We conclude the following points.

1) The security of the proposed TRNG architecture is independent of technology nodes and individual manufacturing differences on chips.

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2) Our design has an excellent performance in terms of robustness against the changes of the supply voltage (0.6–1.8 V) and environmental temperature (−10 °C–40 °C).
3) The generated bitstream without post-processing passes the NIST SP800-22 and Diehard statistical test suites successfully under the above-mentioned physical conditions, and the entropy is not less than 0.999 per bit.
4) An embedded entropy assurance module is necessary in order to assure the high quality of generated bit sequences in practice, e.g., the online entropy estimator and self-adaptive mechanism provided in this work. Our method is oriented to the most basic oscillator structure and has good scalability, so we believe that the proposed on-chip entropy assurance can be used as an elementary component employed in oscillator-based TRNGs. We will further research the applicability of the on-chip entropy assurance to other TRNG designs in the future.

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